Lesson 10: Stochastic Approximation

Robbins–Monro, the ODE Method, and Two-Timescale Analysis

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Curriculum: A Rigorous Learning Path to Multi-Agent Deep RL

Abstract

Stochastic approximation (SA) is the mathematical engine behind virtually every reinforcement learning algorithm. Temporal-Difference learning, Q-learning, SARSA, actor-critic methods, and policy gradient algorithms are all instances of stochastic approximation schemes. This document develops the theory of SA from the classical Robbins–Monro algorithm through the ODE method of Ljung and Kushner–Clark, convergence rate analysis via the central limit theorem for SA, and Borkar's two-timescale framework that underlies actor-critic architectures. We provide complete proofs of all main convergence theorems under precise assumptions, emphasizing the dynamical systems viewpoint: a stochastic approximation iteration with decreasing step sizes asymptotically tracks the solutions of an associated ordinary differential equation. Every major result is connected to specific RL algorithms through detailed worked examples.

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1 Introduction

The central challenge in reinforcement learning is optimization under uncertainty: an agent must find an optimal policy when the environment dynamics are unknown and the only information available comes through noisy, sequential interactions. This is precisely the setting of *stochastic approximation* — finding the root (or optimum) of a function that can only be observed through noisy samples.

The field of stochastic approximation was born with the seminal 1951 paper of Robbins and Monro, who proposed an iterative scheme for finding the root of a regression function from noisy observations. This idea was extended to optimization by Kiefer and Wolfowitz, and the theoretical foundations were deepened by Ljung, Kushner and Clark, and Borkar, who established the powerful *ODE method*: under appropriate conditions, a stochastic approximation algorithm converges to the same limit as the solution of an associated ordinary differential equation.

Why this matters for RL. Consider the following RL update rules:

- **TD(0):** $V(s_n) \leftarrow V(s_n) + \alpha_n\big[R_{n+1} + \gamma V(s_{n+1}) - V(s_n)\big]$
- **Q-learning:** $Q(s_n, a_n) \leftarrow Q(s_n, a_n) + \alpha_n\big[R_{n+1} + \gamma \max_{a'} Q(s_{n+1}, a') - Q(s_n, a_n)\big]$
- **Actor-Critic:** Critic updates value, actor updates policy, at *different rates*
- **REINFORCE:** $\theta_{n+1} = \theta_n + \alpha_n \widehat{\nabla J(\theta_n)}$

Every one of these is a stochastic approximation iteration. The convergence theorems in this document therefore provide the theoretical foundation for understanding *when* and *why* RL algorithms converge.

Document outline. Section 2 introduces the Robbins–Monro algorithm and proves its convergence under classical conditions. Section 3 develops the ODE method, the most powerful general-purpose tool for SA convergence analysis. Section 4 establishes the convergence rate via the CLT for SA and Polyak–Ruppert averaging. Section 5 presents Borkar's two-timescale SA theory, which is the backbone of actor-critic algorithms. Section 6 treats the important special case of linear SA. Section 7 maps the general theory to specific RL algorithms in detail. Section 8 provides exercises, and Section 9 summarizes all key results.

Prerequisites. Probability spaces, conditional expectation, martingale difference sequences, the law of large numbers, and basic ODE theory (existence and uniqueness of solutions, Lyapunov stability).

2 The Robbins–Monro Algorithm

2.1 Problem formulation

Suppose we wish to find a root of a function $h : \mathbb{R}^d \to \mathbb{R}^d$, i.e., a point $x^* \in \mathbb{R}^d$ such that $h(x^*) = 0$. The difficulty is that we cannot evaluate h directly. Instead, for each query point x, we observe a noisy sample

$$H(x, Y) \quad \text{where} \quad h(x) = \mathbb{E}[H(x, Y)]$$

and Y is a random variable (or random element) representing the noise source.

Example 2.1 (Stochastic root-finding)**.** Let $h(\theta) = \mathbb{E}[f(X) - \theta]$ where X is a random variable with unknown distribution. Then $h(\theta^*) = 0$ when $\theta^* = \mathbb{E}[f(X)]$. Given i.i.d. samples $X_1, X_2, \ldots$, the noisy observation is $H(\theta, X_n) = f(X_n) - \theta$.

Definition 2.1 (Step-size sequence)**.** A *step-size sequence* (or *learning rate schedule*) is a sequence $\{a_n\}_{n\geq 0}$ of positive real numbers satisfying:

(i) $\displaystyle\sum_{n=0}^{\infty} a_n = \infty$ (the steps are large enough to reach any target),

(ii) $\displaystyle\sum_{n=0}^{\infty} a_n^2 < \infty$ (the steps are small enough that noise averages out).

Remark 2.1 (Why both conditions are needed). Condition (i) ensures the iterates can travel an unbounded total distance — without it, the algorithm could "freeze" before reaching x^* regardless of the initial condition. Condition (ii) controls the accumulated noise variance: since the noise at step n is scaled by a_n, the total noise variance is proportional to $\sum a_n^2 \mathrm{Var}(\text{noise})$, which must be finite for convergence. The canonical choice is $a_n = c/n$ for some $c > 0$ (since $\sum 1/n = \infty$ while $\sum 1/n^2 < \infty$). More generally, $a_n = c/n^\alpha$ works for any $\alpha \in (1/2, 1]$.

2.2 The iteration

Definition 2.2 (Robbins–Monro iteration)**.** Given an initial point $x_0 \in \mathbb{R}^d$, the *Robbins–Monro iteration* is

$$x_{n+1} = x_n + a_n H(x_n, Y_n), \quad n \geq 0, \tag{1}$$

where $\{a_n\}$ satisfies Definition 2.1 and $Y_0, Y_1, \ldots$ are the noise samples.

We decompose the update into its "signal" and "noise" components. Define the *martingale difference noise*

$$M_{n+1} \coloneqq H(x_n, Y_n) - h(x_n). \tag{2}$$

Then the iteration becomes

$$x_{n+1} = x_n + a_n\big[h(x_n) + M_{n+1}\big]. \tag{3}$$

This is the canonical form we shall use throughout: the iterate moves in the direction $h(x_n)$ (the "drift") plus additive noise M_{n+1}.

2.3 Classical convergence theorem

We state the assumptions precisely and then prove convergence.

Definition 2.3 (Standing assumptions for Robbins–Monro)**.** Let $\mathcal{F}_n = \sigma(x_0, Y_0, \ldots, Y_{n-1})$ be the natural filtration. We assume:

(A1) **Step sizes:** $\{a_n\}$ satisfies Definition 2.1.

(A2) **Martingale difference noise:** $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = 0$ a.s. for all n.

(A3) **Bounded noise variance:** There exists $K > 0$ such that $\mathbb{E}[\|M_{n+1}\|^2 \mid \mathcal{F}_n] \leq K(1+\|x_n\|^2)$ a.s.

(A4) **Stability (monotonicity):** There exists a unique root $x^* \in \mathbb{R}^d$ with $h(x^*) = 0$, and $\langle x - x^*, h(x)\rangle < 0$ for all $x \neq x^*$. (That is, h points "inward" toward x^*.)

(A5) **Linear growth:** $\|h(x)\| \leq K(1 + \|x\|)$ for all x.

Theorem 2.1 (Robbins–Monro convergence)**.** *Under assumptions (A1)–(A5), the Robbins–Monro iteration* (1) *satisfies* $x_n \xrightarrow{a.s.} x^*$.

Proof. We use the Lyapunov function $V(x) = \|x - x^*\|^2$. Define $e_n = x_n - x^*$. Then

$$\begin{aligned}\|e_{n+1}\|^2 &= \|e_n + a_n(h(x_n) + M_{n+1})\|^2 \\ &= \|e_n\|^2 + 2a_n\langle e_n, h(x_n)\rangle + 2a_n\langle e_n, M_{n+1}\rangle \\ &\quad + a_n^2\|h(x_n) + M_{n+1}\|^2.\end{aligned}$$

Taking conditional expectation with respect to $\mathcal{F}_n$ and using (A2):

$$\mathbb{E}[\|e_{n+1}\|^2 \mid \mathcal{F}_n] = \|e_n\|^2 + 2a_n\langle e_n, h(x_n)\rangle + a_n^2\mathbb{E}[\|h(x_n) + M_{n+1}\|^2 \mid \mathcal{F}_n]. \tag{4}$$

For the last term, by the inequality $\|a + b\|^2 \le 2\|a\|^2 + 2\|b\|^2$:

$$\mathbb{E}[\|h(x_n) + M_{n+1}\|^2 \mid \mathcal{F}_n] \le 2\|h(x_n)\|^2 + 2\mathbb{E}[\|M_{n+1}\|^2 \mid \mathcal{F}_n].$$

Using (A3) and (A5):

$$2\|h(x_n)\|^2 + 2\mathbb{E}[\|M_{n+1}\|^2 \mid \mathcal{F}_n] \le 2K^2(1 + \|x_n\|^2) + 2K(1 + \|x_n\|^2) \le C(1 + \|x_n\|^2)$$

for some constant $C > 0$. Since $\|x_n\|^2 \le 2\|e_n\|^2 + 2\|x^*\|^2$, we obtain

$$\mathbb{E}[\|e_{n+1}\|^2 \mid \mathcal{F}_n] \le \|e_n\|^2 + 2a_n\langle e_n, h(x_n)\rangle + a_n^2C'(1 + \|e_n\|^2) \tag{5}$$

for some $C' > 0$.

By (A4), $\langle e_n, h(x_n)\rangle < 0$ when $x_n \ne x^*$. Define $V_n = \|e_n\|^2$ and

$$W_n = \sum_{k=0}^{n-1}(-2a_k)\langle e_k, h(x_k)\rangle, \quad U_n = \sum_{k=0}^{n-1} a_k^2C'(1 + \|e_k\|^2).$$

Then $V_n + W_n - U_n$ is a (nonnegative when adjusted) supermartingale. More precisely, writing $\beta_n = -2\langle e_n, h(x_n)\rangle > 0$ (for $x_n \ne x^*$), we have from (5):

$$\mathbb{E}[V_{n+1} \mid \mathcal{F}_n] \le V_n - a_n\beta_n + a_n^2C'(1 + V_n + 2\|x^*\|^2).$$

We apply the Robbins–Siegmund theorem (a standard supermartingale convergence result).

Robbins–Siegmund Theorem. *Let $\{V_n\}$, $\{\alpha_n\}$, $\{\beta_n\}$, $\{u_n\}$ be nonnegative adapted sequences with $\mathbb{E}[V_{n+1} \mid \mathcal{F}_n] \le (1 + \alpha_n)V_n - \beta_n + u_n$ and $\sum \alpha_n < \infty$, $\sum u_n < \infty$ a.s. Then V_n converges a.s. and $\sum \beta_n < \infty$ a.s.*

In our case, set $\alpha_n = a_n^2C'$, $u_n = a_n^2C'(1 + 2\|x^*\|^2)$, and $\beta_n = a_n(-2\langle e_n, h(x_n)\rangle)$. Since $\sum a_n^2 < \infty$ by (A1), we have $\sum \alpha_n < \infty$ and $\sum u_n < \infty$. By Robbins–Siegmund, $V_n = \|e_n\|^2$ converges a.s. (to some finite limit) and

$$\sum_{n=0}^{\infty} a_n\big(-2\langle e_n, h(x_n)\rangle\big) < \infty \quad \text{a.s.} \tag{6}$$

Suppose for contradiction that $V_n \to L > 0$ with positive probability on some event A. On A, $\|e_n\| \to \sqrt{L} > 0$, so x_n stays bounded away from x^*. By the continuity of h and (A4), there exists $\delta > 0$ such that $-\langle e_n, h(x_n)\rangle \ge \delta$ for all large n on A. Then

$$\sum_n a_n(-2\langle e_n, h(x_n)\rangle) \ge 2\delta \sum_n a_n = \infty,$$

contradicting (6). Therefore $V_n \to 0$ a.s., i.e., $x_n \xrightarrow{\text{a.s.}} x^*$. □

Robbins–Monro Convergence

Under a monotonicity condition on h (it points toward the root x^*), martingale difference noise with bounded conditional variance, and the step-size conditions $\sum a_n = \infty$, $\sum a_n^2 < \infty$, the iteration $x_{n+1} = x_n + a_n[h(x_n) + M_{n+1}]$ converges almost surely to x^*.

SGD as Robbins–Monro

Stochastic gradient descent (SGD) for minimizing $f(\theta) = \mathbb{E}_Z[\ell(\theta, Z)]$ is the Robbins–Monro iteration with $h(\theta) = -\nabla f(\theta)$ and $H(\theta, Z) = -\nabla_\theta \ell(\theta, Z)$. The update is $\theta_{n+1} = \theta_n - a_n \nabla_\theta \ell(\theta_n, Z_n)$. Under convexity of f (which implies the monotonicity condition (A4) for $-\nabla f$), Theorem 2.1 guarantees $\theta_n \xrightarrow{\text{a.s.}} \theta^*$.

3 The ODE Method

The Robbins–Monro analysis requires the strong monotonicity condition (A4), which fails for many RL algorithms. The *ODE method*, developed by Ljung (1977) and Kushner–Clark (1978), replaces this with a more natural condition: convergence of the SA iteration follows from the stability of the associated ODE. This is the primary tool for analyzing RL convergence.

3.1 The associated ODE

Definition 3.1 (Associated ODE)**.** Given the SA iteration (3), the *associated ODE* is

$$\dot{x}(t) = h(x(t)), \quad x(0) = x_0. \tag{7}$$

The key insight is that for small step sizes, the SA iteration $x_{n+1} \approx x_n + a_n h(x_n)$ behaves like an Euler discretization of the ODE (7) with time step a_n, plus noise that averages out.

3.2 Interpolated trajectory

To make the connection between the discrete SA iteration and the continuous ODE precise, we construct an interpolated trajectory.

Definition 3.2 (Interpolated trajectory)**.** Define the time instants $t_0 = 0$ and $t_n = \sum_{k=0}^{n-1} a_k$ for $n \geq 1$. Note that $t_n \to \infty$ by condition (i) of Definition 2.1. The *interpolated trajectory* $\bar{x} : [0, \infty) \to \mathbb{R}^d$ is the piecewise constant function

$$\bar{x}(t) = x_n \quad \text{for } t \in [t_n, t_{n+1}).$$

Equivalently, one may use the piecewise linear interpolation; both lead to the same asymptotic analysis.

3.3 Assumptions for the ODE method

Definition 3.3 (ODE method assumptions)**.** We strengthen and modify the assumptions as follows:

(B1) **Step sizes:** As in (A1).

(B2) **Lipschitz drift:** $h : \mathbb{R}^d \to \mathbb{R}^d$ is Lipschitz continuous: $\|h(x) - h(y)\| \leq L\|x - y\|$ for all x, y and some $L > 0$.

(B3) **Martingale difference noise:** $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = 0$ a.s. and $\mathbb{E}[\|M_{n+1}\|^2 \mid \mathcal{F}_n] \leq K(1 + \|x_n\|^2)$ a.s.

(B4) **Stability (boundedness):** $\sup_n \|x_n\| < \infty$ a.s.

(B5) **Asymptotic stability of the ODE:** The ODE (7) has a globally asymptotically stable equilibrium x^* (i.e., $h(x^*) = 0$ and $x(t) \to x^*$ as $t \to \infty$ for every initial condition $x(0) \in \mathbb{R}^d$).

Remark 3.1 (Stability assumption (B4)). Assumption (B4) — that the iterates remain bounded — is the most delicate. In practice, one either (a) proves it from properties of h (see the Borkar–Meyn theorem in Section 3.6), (b) uses projection to a compact set, or (c) verifies it empirically. For many RL algorithms, boundedness follows from the structure of the MDP (e.g., bounded rewards and a discount factor $\gamma < 1$).

3.4 The noise vanishes asymptotically

The first step in the ODE method is to show that the accumulated noise becomes negligible.

Lemma 3.1 (Noise term vanishes). *Under (B1), (B3), and (B4), the process $Z_n = \sum_{k=0}^{n-1} a_k M_{k+1}$ converges a.s. In particular, for any $T > 0$ and $\varepsilon > 0$,*

$$\sup_{t_n \le s \le t_n + T} \Big\| \sum_{k=n}^{m(s)-1} a_k M_{k+1} \Big\| \to 0 \quad \textit{a.s. as } n \to \infty,$$

where $m(s) = \min\{j : t_j > s\}$.

Proof. The sequence $Z_n = \sum_{k=0}^{n-1} a_k M_{k+1}$ is a martingale with respect to $\mathcal{F}_n$. Its quadratic variation satisfies

$$\mathbb{E}[\|Z_n\|^2] = \sum_{k=0}^{n-1} a_k^2 \mathbb{E}[\|M_{k+1}\|^2] \le \sum_{k=0}^{n-1} a_k^2 K(1 + \mathbb{E}[\|x_k\|^2]).$$

By (B4), $\|x_k\|$ is bounded a.s., so $\mathbb{E}[\|x_k\|^2] \le C$ for some $C > 0$. Thus $\mathbb{E}[\|Z_n\|^2] \le K(1 + C) \sum_{k=0}^{\infty} a_k^2 < \infty$ by (B1). By the martingale convergence theorem, Z_n converges a.s.

For the uniform convergence over windows of length T, note that for $t_n \le s \le t_n + T$, the "tail sum" $\sum_{k=n}^{m(s)-1} a_k M_{k+1}$ is a portion of the convergent series $\sum a_k M_{k+1}$, so it tends to zero uniformly over bounded time windows. More precisely, since Z_n converges a.s., $\|Z_m - Z_n\| \to 0$ as $n, m \to \infty$, and the number of indices k with $t_n \le t_k \le t_n + T$ increases but $a_k \to 0$, giving the claimed uniform convergence. □

3.5 Main convergence theorem

Theorem 3.2 (Convergence via the ODE method). *Under assumptions (B1)–(B5), the SA iteration $x_{n+1} = x_n + a_n[h(x_n) + M_{n+1}]$ satisfies $x_n \xrightarrow{a.s.} x^*$.*

Proof. The proof proceeds in three steps.

Step 1: Shifted interpolated trajectories. For each $n \ge 0$, define the shifted interpolated trajectory $x^{(n)} : [0, \infty) \to \mathbb{R}^d$ by

$$x^{(n)}(0) = x_n, \quad x^{(n)}(t) = x_{n+k} \text{ for } t \in [t_{n+k} - t_n, t_{n+k+1} - t_n), \quad k \ge 0.$$

This is the piecewise constant interpolation of the SA iterates starting from time t_n, with the time axis reset to zero.

Step 2: Equicontinuity and convergence to ODE solutions. By (B4), the iterates $\{x_n\}$ are bounded a.s. The drift h is Lipschitz by (B2), so $\|h(x_n)\| \le L\|x_n\| + \|h(0)\|$ is bounded. For any $T > 0$ and $s, t \in [0, T]$ with $s < t$, the trajectory satisfies

$$x^{(n)}(t) - x^{(n)}(s) = \sum_{k: t_k \in [s+t_n, t+t_n)} a_k[h(x_k) + M_{k+1}].$$

The "signal" part is bounded by $\|h\|_\infty \cdot (t - s)$ (since $\sum a_k \approx t - s$ over the corresponding indices). By Lemma 3.1, the noise part vanishes as $n \to \infty$. Therefore, the family $\{x^{(n)}\}$ is equicontinuous on $[0, T]$ a.s. for large n.

By the Arzela–Ascoli theorem, every subsequence of $\{x^{(n)}\}$ has a further subsequence that converges uniformly on $[0, T]$. Let $x^{(\infty)}$ be any such limit. We claim $x^{(\infty)}$ satisfies the ODE (7).

Indeed, the SA iteration gives

$$x^{(n)}(t) = x^{(n)}(0) + \int_0^t h(x^{(n)}(s))\, ds + \varepsilon_n(t),$$

where $\varepsilon_n(t)$ accounts for the discretization error and the noise, both of which vanish as $n \to \infty$ (the discretization error vanishes because $a_n \to 0$ by (B1), and the noise vanishes by Lemma 3.1). Taking the limit and using the Lipschitz continuity of h to pass through the integral:

$$x^{(\infty)}(t) = x^{(\infty)}(0) + \int_0^t h(x^{(\infty)}(s))\, ds.$$

Thus $x^{(\infty)}$ is a solution of the ODE (7).

Step 3: Convergence to the equilibrium. By (B5), the equilibrium x^* is globally asymptotically stable for the ODE. Therefore, for any solution $x(t)$ of the ODE, $x(t) \to x^*$ as $t \to \infty$.

Now we argue that $x_n \to x^*$ a.s. The set of limit points of $\{x_n\}$ is nonempty (by boundedness (B4)) and compact. From Step 2, any limit point of the discrete sequence is also an initial condition for an ODE trajectory that must converge to x^*. Moreover, the set of limit points is connected (as the image of a connected set under a continuous map) and invariant under the ODE flow.

The only compact, connected, ODE-flow-invariant set that is attracted to x^* under a globally asymptotically stable system is $\{x^*\}$ itself. To see this, suppose $\bar{x} \ne x^*$ is a limit point. Then there exists a subsequence $x_{n_k} \to \bar{x}$. The shifted trajectory $x^{(n_k)}$ converges to an ODE solution starting at $\bar{x}$, which satisfies $x(T) \to x^*$ for large T. But then the iterates after n_k must eventually be close to x^*, and by the same argument, they cannot leave a neighborhood of x^* (by Lyapunov stability of x^*). Hence x^* is the unique limit point, and $x_n \xrightarrow{\text{a.s.}} x^*$. □

ODE Method – Main Principle

A stochastic approximation iteration $x_{n+1} = x_n + a_n[h(x_n) + M_{n+1}]$ with decreasing step sizes converges to the set of asymptotically stable equilibria of the ODE $\dot{x} = h(x)$, provided the iterates remain bounded and the noise is a martingale difference sequence.

3.6 The Borkar–Meyn stability theorem

Assumption (B4) (boundedness of iterates) is often the hardest to verify. The Borkar–Meyn theorem provides sufficient conditions.

Definition 3.4 (Borkar–Meyn assumptions)**.** In addition to (B1)–(B3), assume:

(C1) h is Lipschitz and the "scaled" function $h_\infty(x) := \lim_{r\to\infty} h(rx)/r$ exists uniformly on compact sets.

(C2) The ODE $\dot{x} = h_\infty(x)$ has the origin as its unique globally asymptotically stable equilibrium.

Theorem 3.3 (Borkar–Meyn stability)**.** *Under (B1)–(B3) and (C1)–(C2), the iterates $\{x_n\}$ are bounded almost surely:* $\sup_n \|x_n\| < \infty$ *a.s.*

Proof. We give the key steps of the proof.

Step 1: Rescaling. Suppose, for contradiction, that $\|x_n\| \to \infty$ along a subsequence with positive probability. Let $r_n = \max(\|x_0\|, \|x_1\|, \ldots, \|x_n\|)$ and consider the rescaled iterates $\hat{x}_n = x_n/r_n$.

Step 2: Rescaled iteration. Since $x_{n+1} = x_n + a_n[h(x_n) + M_{n+1}]$, dividing by r_{n+1}:

$$\hat{x}_{n+1} = \frac{r_n}{r_{n+1}}\hat{x}_n + \frac{a_n}{r_{n+1}}[h(r_n\hat{x}_n) + M_{n+1}].$$

When $r_n \to \infty$, $h(r_n\hat{x}_n)/r_n \to h_\infty(\hat{x}_n)$ by (C1), and $r_n/r_{n+1} \to 1$ since $a_n \to 0$. The noise term $a_n M_{n+1}/r_{n+1}$ becomes negligible.

Step 3: Limiting ODE. By the same interpolation argument as in Theorem 3.2, the rescaled trajectory converges (along subsequences) to solutions of the ODE $\dot{x} = h_\infty(x)$. But $\|\hat{x}_n\| \le 1$ and $\|\hat{x}_n\| = 1$ infinitely often (when r_n increases). So the limiting trajectory is confined to the unit ball and visits its boundary.

Step 4: Contradiction. By (C2), every solution of $\dot{x} = h_\infty(x)$ converges to the origin. But the limiting trajectory of $\hat{x}_n$ must repeatedly visit $\|\hat{x}\| = 1$, contradicting the global asymptotic stability of the origin for the h_∞ ODE. Therefore $\sup_n \|x_n\| < \infty$ a.s. □

Remark 3.2 (Practical significance). The Borkar–Meyn theorem is crucial because it converts the verification of iterate boundedness into a condition on the "behavior of h at infinity." For linear SA where $h(x) = Ax + b$, we have $h_\infty(x) = Ax$, so (C2) reduces to requiring that A is Hurwitz (all eigenvalues have negative real parts). This is exactly the condition for TD(0) convergence with linear function approximation.

4 Convergence Rate Analysis

Having established convergence, we now ask: *how fast* does $x_n \to x^*$?

4.1 Central limit theorem for SA

Definition 4.1 (Assumptions for the CLT)**.** Assume (B1)–(B5) hold, and additionally:

(D1) h is differentiable at x^* with Jacobian $A = Dh(x^*) = \nabla h(x^*) \in \mathbb{R}^{d\times d}$.

(D2) A is Hurwitz: all eigenvalues of A have strictly negative real parts. (This is automatic from the asymptotic stability of x^* for the linearized ODE.)

(D3) The step size is $a_n = a/n$ for some constant $a > 0$ such that $aA + \frac{1}{2}I$ is Hurwitz (i.e., all eigenvalues of aA have real part less than $-1/2$).

(D4) The conditional covariance converges: $\mathbb{E}[M_{n+1}M_{n+1}^T \mid \mathcal{F}_n] \to \Sigma$ in probability as $x_n \to x^*$, for some positive definite matrix Σ.

Theorem 4.1 (CLT for stochastic approximation)**.** *Under the assumptions of Definition 4.1, we have*

$$\sqrt{n}(x_n - x^*) \xrightarrow{d} \mathcal{N}(0, V),$$

where V is the unique solution of the Lyapunov equation

$$(aA + \tfrac{1}{2}I)V + V(aA + \tfrac{1}{2}I)^T + a^2\Sigma = 0. \tag{8}$$

Proof. Define $e_n = x_n - x^*$ and $u_n = \sqrt{n}\, e_n$. Near x^*, the drift is approximately linear:

$$h(x_n) = h(x^*) + Ae_n + o(\|e_n\|) = Ae_n + o(\|e_n\|),$$

since $h(x^*) = 0$. With $a_n = a/n$, the SA iteration gives

$$e_{n+1} = e_n + \frac{a}{n}[Ae_n + r_n + M_{n+1}],$$

where $r_n = h(x_n) - Ae_n = o(\|e_n\|)$ is the higher-order remainder.

Substituting $e_n = u_n/\sqrt{n}$:

$$\frac{u_{n+1}}{\sqrt{n+1}} = \frac{u_n}{\sqrt{n}} + \frac{a}{n}\Big[A\frac{u_n}{\sqrt{n}} + r_n + M_{n+1}\Big].$$

Multiplying both sides by $\sqrt{n+1}$ and using $\sqrt{n+1}/\sqrt{n} = 1 + 1/(2n) + O(1/n^2)$:

$$\begin{aligned} u_{n+1} &= \sqrt{\frac{n+1}{n}}\, u_n + \frac{a\sqrt{n+1}}{n}\Big[A\frac{u_n}{\sqrt{n}} + r_n + M_{n+1}\Big] \\ &= \Big(1 + \frac{1}{2n} + O(1/n^2)\Big)u_n + \frac{a}{n}\Big(1 + O(1/n)\Big)[Au_n + \sqrt{n}\, r_n + \sqrt{n}\, M_{n+1}] \\ &= u_n + \frac{1}{n}\Big(aA + \frac{1}{2}I\Big)u_n + \frac{a}{\sqrt{n}}M_{n+1} + \text{negligible terms.} \end{aligned} \tag{9}$$

The term $\sqrt{n}\, r_n/n = r_n/\sqrt{n}$ vanishes because $r_n = o(\|e_n\|) = o(\|u_n\|/\sqrt{n})$ and $u_n = O(1)$ in the limit.

The recursion (9) is itself a stochastic approximation with step size $1/n$, drift $g(u) = (aA + \frac{1}{2}I)u$, and diffusion coefficient $a/\sqrt{n} \cdot \sqrt{n} = a$ times the noise M_{n+1}. By (D3), the matrix $aA + \frac{1}{2}I$ is Hurwitz, so the origin is a stable equilibrium of the drift. The process $\{u_n\}$ is a linear stochastic recursion whose stationary distribution is Gaussian.

Standard results on such recursions (or a direct characteristic function argument) yield $u_n \xrightarrow{d} \mathcal{N}(0, V)$ where V satisfies the Lyapunov equation (8). Specifically, the covariance $V_n = \mathbb{E}[u_n u_n^T]$ satisfies

$$V_{n+1} \approx V_n + \frac{1}{n}\Big[(aA + \tfrac{1}{2}I)V_n + V_n(aA + \tfrac{1}{2}I)^T + a^2\Sigma\Big],$$

and the fixed point $V_{n+1} = V_n = V$ gives (8). □

Remark 4.1 (Optimal step size). The asymptotic covariance V depends on a through the Lyapunov equation (8). When $d = 1$ and $A = -\lambda$ with $\lambda > 0$, the equation becomes $(-2a\lambda + 1)V = a^2\sigma^2$, giving $V = a^2\sigma^2/(2a\lambda - 1)$. This is minimized when $a = 1/\lambda$, yielding $V_{\text{opt}} = \sigma^2/\lambda^2$. In higher dimensions, the optimal choice requires $a = -(A^T)^{-1}/2$, which depends on unknown quantities.

4.2 Polyak–Ruppert averaging

The optimal step size depends on the Jacobian A at x^*, which is typically unknown. Polyak (1990) and Ruppert (1988) independently proposed a simple device that achieves the optimal rate *without* knowing A: average the iterates.

Theorem 4.2 (Polyak–Ruppert averaging)**.** *Let $\{x_n\}$ be the SA iterates with step size $a_n = a/n^\alpha$ for $\alpha \in (1/2, 1)$, and define the averaged iterate*

$$\bar{x}_n = \frac{1}{n}\sum_{k=1}^{n} x_k.$$

Under assumptions (B1)–(B5) and (D1)–(D4) (with the modified step size), we have

$$\sqrt{n}(\bar{x}_n - x^*) \xrightarrow{d} \mathcal{N}(0, A^{-1}\Sigma(A^{-T})).$$

This is the optimal asymptotic covariance (in the sense of the Cramer–Rao bound for the root-finding problem).

Proof. Since $x_n \to x^*$ and $h(x^*) = 0$, we have

$$x_{n+1} - x_n = a_n[h(x_n) + M_{n+1}] = a_n[A(x_n - x^*) + r_n + M_{n+1}],$$

where $r_n = o(\|x_n - x^*\|)$. Summing from $k = 1$ to n:

$$x_{n+1} - x_1 = A\sum_{k=1}^{n} a_k(x_k - x^*) + \sum_{k=1}^{n} a_k r_k + \sum_{k=1}^{n} a_k M_{k+1}.$$

Rearranging:

$$A\sum_{k=1}^{n} a_k(x_k - x^*) = (x_{n+1} - x_1) - \sum_{k=1}^{n} a_k r_k - \sum_{k=1}^{n} a_k M_{k+1}.$$

Define $S_n = \sum_{k=1}^{n} a_k$. Since $a_k = a/k^\alpha$ with $\alpha < 1$, $S_n \sim \frac{a}{1-\alpha} n^{1-\alpha}$.

Now, the key observation: $\bar{x}_n - x^* = \frac{1}{n}\sum_{k=1}^{n}(x_k - x^*)$. We relate this to the weighted sum above. A careful Abel summation argument shows that

$$\sqrt{n}(\bar{x}_n - x^*) = -A^{-1} \cdot \frac{1}{\sqrt{n}} \sum_{k=1}^{n} M_{k+1} + o_P(1).$$

The dominant term is the sample average of the martingale differences, and the remainder terms ($x_{n+1} - x_1$ is $o(\sqrt{n})$, and $\sum a_k r_k$ is negligible) vanish after normalization.

By the martingale CLT, $\frac{1}{\sqrt{n}}\sum_{k=1}^{n} M_{k+1} \xrightarrow{d} \mathcal{N}(0, \Sigma)$. Therefore

$$\sqrt{n}(\bar{x}_n - x^*) \xrightarrow{d} \mathcal{N}(0, A^{-1}\Sigma A^{-T}).$$

□

Polyak–Ruppert Averaging

By averaging the iterates of a slowly decaying SA scheme ($a_n = O(n^{-\alpha})$, $\alpha \in (1/2, 1)$), the averaged iterate $\bar{x}_n$ achieves the optimal convergence rate $O(1/\sqrt{n})$ with the minimal asymptotic covariance $A^{-1}\Sigma A^{-T}$, without knowledge of the Jacobian $A = Dh(x^*)$.

Iterate Averaging in RL

Polyak–Ruppert averaging suggests that RL algorithms should average their iterates rather than use the last iterate. In practice, *exponential moving averages* (EMA) are used as a practical approximation: the target network in DQN, the exponentially averaged value function in some actor-critic methods, and the "Polyak averaging" of policy parameters in DDPG/TD3 all implement this principle. The theoretical guarantee is that averaging provably improves the convergence rate.

5 Two-Timescale Stochastic Approximation

5.1 Motivation

Many RL algorithms involve two coupled iterative updates running simultaneously but at different rates. The canonical example is *actor-critic*: the critic updates the value function (fast timescale) while the actor updates the policy (slow timescale). The theory of two-timescale SA, developed by Borkar (1997), provides the convergence framework for such algorithms.

5.2 Formulation

Definition 5.1 (Two-timescale SA)**.** Consider the coupled iterations

$$x_{n+1} = x_n + a_n[f(x_n, y_n) + M_{n+1}^{(1)}], \tag{10}$$

$$y_{n+1} = y_n + b_n[g(x_n, y_n) + M_{n+1}^{(2)}], \tag{11}$$

where $x_n \in \mathbb{R}^{d_1}$, $y_n \in \mathbb{R}^{d_2}$, and the step-size sequences satisfy:

(i) $\sum a_n = \sum b_n = \infty$ and $\sum a_n^2 < \infty$, $\sum b_n^2 < \infty$,

(ii) $a_n/b_n \to 0$ as $n \to \infty$ (the "timescale separation").

The typical choice is $a_n = 1/n$ and $b_n = 1/n^\beta$ for $\beta \in (1/2, 1)$, or $a_n = 1/n$ and $b_n = 1/n^{2/3}$.

Remark 5.1 (Timescale separation). The condition $a_n/b_n \to 0$ means that the y-iteration uses *larger* step sizes than the x-iteration, so y moves faster. From the perspective of the y-update, x changes so slowly that it appears quasi-static. From the perspective of the x-update, y equilibrates so quickly that it appears to have already converged.

5.3 Assumptions

Definition 5.2 (Two-timescale assumptions)**.** (E1) **Step sizes:** As in Definition 5.1.

(E2) **Lipschitz drifts:** $f : \mathbb{R}^{d_1} \times \mathbb{R}^{d_2} \to \mathbb{R}^{d_1}$ and $g : \mathbb{R}^{d_1} \times \mathbb{R}^{d_2} \to \mathbb{R}^{d_2}$ are globally Lipschitz.

(E3) **Martingale difference noise:** $\mathbb{E}[M_{n+1}^{(i)} \mid \mathcal{F}_n] = 0$ and $\mathbb{E}[\|M_{n+1}^{(i)}\|^2 \mid \mathcal{F}_n] \le K(1 + \|x_n\|^2 + \|y_n\|^2)$ for $i = 1, 2$.

(E4) **Stability:** $\sup_n(\|x_n\| + \|y_n\|) < \infty$ a.s.

(E5) **Fast timescale equilibrium:** For each fixed $x \in \mathbb{R}^{d_1}$, the ODE $\dot{y} = g(x, y)$ has a globally asymptotically stable equilibrium $\lambda(x)$, and $\lambda : \mathbb{R}^{d_1} \to \mathbb{R}^{d_2}$ is Lipschitz.

(E6) **Slow timescale stability:** The ODE $\dot{x} = f(x, \lambda(x))$ has a globally asymptotically stable equilibrium x^*.

5.4 Convergence theorem

Theorem 5.1 (Two-timescale convergence (Borkar))**.** *Under assumptions (E1)–(E6), the two-timescale SA iterates satisfy*

$$x_n \xrightarrow{a.s.} x^*, \quad y_n \xrightarrow{a.s.} \lambda(x^*) := y^*.$$

Proof. The proof is structured in two phases, analyzing each timescale separately.

Phase 1: Fast timescale analysis (convergence of y_n).

Consider the y-iteration (11). Since $a_n/b_n \to 0$, the x-iterates change negligibly on the timescale of b_n. Formally, on the time axis defined by $\tau_n = \sum_{k=0}^{n-1} b_k$, the x-update has effective step size a_n, which is asymptotically negligible compared to b_n.

To make this precise, fix any $T > 0$ and consider the window of indices $[n, m_n]$ where m_n is the largest index such that $\sum_{k=n}^{m_n-1} b_k \le T$. On this window:

$$\|x_{m_n} - x_n\| \le \sum_{k=n}^{m_n-1} a_k \|f(x_k, y_k) + M_{k+1}^{(1)}\| \le C \sum_{k=n}^{m_n-1} a_k,$$

where C is a bound on $\|f\| + \|M^{(1)}\|$ (which exists by (E4) and (E2)). Now

$$\sum_{k=n}^{m_n-1} a_k = \sum_{k=n}^{m_n-1} \frac{a_k}{b_k} \cdot b_k \leq \sup_{k\geq n} \frac{a_k}{b_k} \sum_{k=n}^{m_n-1} b_k \leq T \cdot \sup_{k\geq n} \frac{a_k}{b_k} \to 0.$$

So on the fast timescale, x is indeed quasi-static.

Therefore, the y-iteration behaves asymptotically like

$$y_{n+1} = y_n + b_n[g(\bar{x}, y_n) + M_{n+1}^{(2)}]$$

for a "frozen" value $\bar{x}$ (which equals the current value of the slowly drifting x_n). By the ODE method (Theorem 3.2), this converges to the equilibrium $\lambda(\bar{x})$ of $\dot{y} = g(\bar{x}, y)$.

More precisely, the y-iterates asymptotically track the ODE $\dot{y} = g(x(t), y)$ where $x(t)$ is the slowly varying x-component. Since x changes slowly, the y-component "quasi-statically" tracks $\lambda(x_n)$: for any $\varepsilon > 0$,

$$\|y_n - \lambda(x_n)\| < \varepsilon \quad \text{for all large } n, \text{ a.s.} \tag{12}$$

Phase 2: Slow timescale analysis (convergence of x_n).

Now consider the x-iteration (10). By (12), $y_n \approx \lambda(x_n)$ for large n, so

$$f(x_n, y_n) = f(x_n, \lambda(x_n)) + [f(x_n, y_n) - f(x_n, \lambda(x_n))].$$

The second term is bounded by $L_f\|y_n - \lambda(x_n)\| \to 0$ a.s. by the Lipschitz property of f.

Therefore, the x-iteration asymptotically behaves like

$$x_{n+1} = x_n + a_n[f(x_n, \lambda(x_n)) + \tilde{M}_{n+1}],$$

where $\tilde{M}_{n+1}$ absorbs the original noise $M_{n+1}^{(1)}$ and the approximation error from replacing y_n by $\lambda(x_n)$. The associated ODE is

$$\dot{x} = f(x, \lambda(x)),$$

which has x^* as a globally asymptotically stable equilibrium by (E6). Applying the ODE method again, $x_n \xrightarrow{\text{a.s.}} x^*$.

Since λ is continuous and $x_n \to x^*$, we conclude $y_n \to \lambda(x^*) = y^*$ a.s. □

Two-Timescale Convergence Principle

In a two-timescale SA system where $a_n/b_n \to 0$:

1. The fast iterate y_n "tracks" the equilibrium map $\lambda(x_n)$ where $g(x, \lambda(x)) = 0$.
2. The slow iterate x_n converges as if y_n were always at equilibrium, driven by the averaged dynamics $\dot{x} = f(x, \lambda(x))$.
3. The overall system converges to $(x^*, y^*) = (x^*, \lambda(x^*))$.

Actor-Critic as Two-Timescale SA

In actor-critic methods, the critic updates the value function estimate w_n on the fast timescale and the actor updates the policy parameters θ_n on the slow timescale:

$$w_{n+1} = w_n + b_n[\delta_n \phi(s_n) - (\text{critic update})],$$
$$\theta_{n+1} = \theta_n + a_n[\psi_n - (\text{actor update})],$$

with $a_n/b_n \to 0$. Here $(x_n, y_n) = (\theta_n, w_n)$, f is the policy gradient direction, and g is the TD error for the current policy. The fast timescale equilibrium $\lambda(\theta)$ is the true

value function V^{π_θ} for policy π_θ. The slow timescale ODE is the policy gradient flow $\dot{\theta} = \nabla J(\theta)$. Theorem 5.1 then guarantees convergence to a local optimum of $J(\theta)$ (or a global optimum if J is concave in θ).

5.5 Lock-in probability

When the ODE has multiple locally stable equilibria (as is typical for nonlinear systems), the SA iteration converges to one of them. Which one depends on the initial condition and the noise history.

Proposition 5.2 (Lock-in to attractors)**.** *Suppose the ODE $\dot{x} = h(x)$ has finitely many equilibria, each of which is either asymptotically stable or unstable. Under the ODE method assumptions (B1)–(B4), the SA iterates converge a.s. to one of the stable equilibria. The probability of converging to an unstable equilibrium is zero.*

Proof. By the ODE method analysis (Theorem 3.2, Step 2), the limit set of $\{x_n\}$ is a compact connected invariant set of the ODE flow.

If this limit set contained an unstable equilibrium $\bar{x}$, then since $\bar{x}$ is unstable, there exists an open neighborhood U of $\bar{x}$ and a direction along which ODE trajectories leave U exponentially fast. However, the noise in SA prevents the iterates from remaining exactly on the unstable manifold.

Formally, by the "avoidance of unstable equilibria" result (Pemantle, 1990; Brandiere, 1998): under the martingale noise assumption with nondegenerate conditional covariance (i.e., $\mathbb{E}[M_{n+1}M_{n+1}^T \mid \mathcal{F}_n]$ is positive definite), the set of initial conditions x_0 for which x_n converges to an unstable equilibrium has Lebesgue measure zero, and $\mathbb{P}(x_n \to \bar{x}) = 0$ for any fixed x_0.

The key idea: near an unstable equilibrium, the noise keeps "kicking" the iterate off the unstable manifold, and once it leaves a small neighborhood, the ODE dynamics carry it toward a stable equilibrium. □

Multiple Stable Equilibria in RL

When the mean-field ODE has multiple stable equilibria, the RL algorithm may converge to a suboptimal one depending on the initial conditions and exploration noise. This is a fundamental challenge in policy gradient methods, where the objective $J(\theta)$ is generally non-concave and has multiple local optima. Two-timescale SA (actor-critic) inherits this issue. Entropy regularization (as in SAC) and exploration bonuses are practical techniques to mitigate convergence to poor local optima.

6 Linear Stochastic Approximation

The most important special case for RL is *linear* SA, where the drift function is affine.

6.1 Setup

Definition 6.1 (Linear SA)**.** A *linear stochastic approximation* iteration is

$$x_{n+1} = x_n + a_n[Ax_n + b + M_{n+1}], \tag{13}$$

where $A \in \mathbb{R}^{d\times d}$, $b \in \mathbb{R}^d$, and $h(x) = Ax + b$.

The associated ODE is $\dot{x} = Ax + b$, a linear ODE with equilibrium $x^* = -A^{-1}b$ (when A is invertible).

6.2 Convergence conditions

Theorem 6.1 (Convergence of linear SA)**.** *Suppose A is Hurwitz (all eigenvalues have strictly negative real parts), the step sizes satisfy (A1), the noise satisfies (B3), and the iterates are bounded a.s. Then $x_n \xrightarrow{a.s.} x^* = -A^{-1}b$.*

Proof. The ODE $\dot{x} = Ax + b$ has $x^* = -A^{-1}b$ as its unique equilibrium. Since A is Hurwitz, the function $V(x) = (x - x^*)^T P(x - x^*)$ is a Lyapunov function for the ODE, where P is the unique positive definite solution of the Lyapunov matrix equation $A^T P + PA = -I$. (Such P exists precisely because A is Hurwitz.)

Indeed,

$$\dot{V} = 2(x-x^*)^T P(Ax+b) = 2(x-x^*)^T PA(x-x^*) = (x-x^*)^T(A^T P+PA)(x-x^*) = -\|x-x^*\|^2,$$

so V is strictly decreasing along nonconstant trajectories. This confirms that x^* is globally asymptotically stable.

The result then follows directly from the ODE method (Theorem 3.2). □

Remark 6.1 (Hurwitz condition for linear SA). The Hurwitz condition on A means that the "drift" $h(x) = Ax + b$ always pushes the iterate toward the equilibrium $x^* = -A^{-1}b$. If any eigenvalue of A had a positive real part, the ODE would be unstable in the corresponding direction, and the SA iteration would diverge. For the borderline case (eigenvalues on the imaginary axis), convergence depends on higher-order terms and the noise structure.

6.3 Stability via Borkar–Meyn for linear SA

Corollary 6.2 (Automatic stability for linear SA)**.** *For linear SA (13) with A Hurwitz, the boundedness assumption $\sup_n \|x_n\| < \infty$ a.s. holds automatically (without needing to assume it).*

Proof. Apply the Borkar–Meyn theorem (Theorem 3.3). For linear $h(x) = Ax + b$, the "function at infinity" is $h_\infty(x) = \lim_{r\to\infty}(A(rx)+b)/r = Ax$. Since A is Hurwitz, the ODE $\dot{x} = Ax$ has the origin as its globally asymptotically stable equilibrium. Thus (C2) is satisfied, and Theorem 3.3 gives $\sup_n \|x_n\| < \infty$ a.s. □

Theorem 6.3 (Complete convergence for linear SA)**.** *For the linear SA iteration (13) with A Hurwitz, step sizes satisfying (A1), and noise satisfying (B3), we have $x_n \xrightarrow{a.s.} -A^{-1}b$ without any additional stability assumptions.*

Proof. Combine Corollary 6.2 (which gives boundedness) with Theorem 6.1 (which gives convergence given boundedness). □

TD(0) with Linear Function Approximation

In TD(0) with linear function approximation $V(s) = \phi(s)^T w$, the parameter vector w is updated by

$$w_{n+1} = w_n + \alpha_n\big[R_{n+1} + \gamma\phi(s_{n+1})^T w_n - \phi(s_n)^T w_n\big]\phi(s_n).$$

This is linear SA with

$$A = \mathbb{E}_\pi[\phi(s)(\gamma\phi(s') - \phi(s))^T], \quad b = \mathbb{E}_\pi[R\phi(s)],$$

where the expectation is over the stationary distribution of the Markov chain under policy π. The key question is: *is A Hurwitz?*

Under on-policy sampling (the states are visited according to the stationary distribution d^π of π), it can be shown that A is negative definite (hence Hurwitz). Specifically, $A =$

$\Phi^T D(\gamma P^\pi - I)\Phi$ where D is the diagonal matrix of stationary probabilities, and $\gamma P^\pi - I$ is negative definite on the range of Φ (since $\gamma < 1$ and P^π has spectral radius 1).
By Theorem 6.3, TD(0) with linear function approximation converges a.s. under on-policy sampling. This is the celebrated result of Tsitsiklis and Van Roy (1997).
The Deadly Triad: Under *off-policy* sampling with function approximation, A may fail to be Hurwitz, and TD(0) can *diverge*. This is the theoretical explanation for the deadly triad (function approximation + bootstrapping + off-policy).

7 Applications to Reinforcement Learning

We now systematically map the general SA theory to specific RL algorithms, verifying the assumptions of the convergence theorems in each case.

7.1 TD(0) as stochastic approximation

TD(0) – Complete SA Analysis

Setting. Consider a finite MDP with states $\mathcal{S} = \{1, \dots, S\}$, policy π, discount factor $\gamma \in [0, 1)$, transition matrix P^π, and reward function $r^\pi(s) = \mathbb{E}[R \mid S = s, \pi]$.
The update. TD(0) updates the value function estimate V by

$$V_{n+1}(s_n) = V_n(s_n) + \alpha_n(s_n)\underbrace{\big[R_{n+1} + \gamma V_n(s_{n+1}) - V_n(s_n)\big]}_{\delta_n \text{ (TD error)}}.$$

SA form. Viewing $V \in \mathbb{R}^S$ as the iterate, define

$$h(V) = r^\pi + \gamma P^\pi V - V = (\gamma P^\pi - I)V + r^\pi.$$

Then $h(V^\pi) = r^\pi + \gamma P^\pi V^\pi - V^\pi = 0$ by the Bellman equation. The noisy observation at step n (for state s_n) is $H_n = (R_{n+1} + \gamma V_n(s_{n+1}) - V_n(s_n))e_{s_n}$, where e_s is the s-th standard basis vector.
Mean field. $\mathbb{E}[H_n \mid V_n, s_n] = (r^\pi(s_n) + \gamma(P^\pi V_n)(s_n) - V_n(s_n))e_{s_n}$, which matches $h(V_n)$ projected onto the s_n-coordinate.
Associated ODE. $\dot{V} = (\gamma P^\pi - I)V + r^\pi$. The matrix $A = \gamma P^\pi - I$ has eigenvalues $\gamma\lambda_i - 1$ where λ_i are eigenvalues of P^π. Since $|\lambda_i| \le 1$ (stochastic matrix) and $\gamma < 1$, all eigenvalues of A satisfy $\mathrm{Re}(\gamma\lambda_i - 1) \le \gamma - 1 < 0$. Hence A is Hurwitz.
Conclusion. By Theorem 6.3 (or Theorem 3.2), TD(0) converges a.s. to $V^* = -A^{-1}r^\pi = (I - \gamma P^\pi)^{-1}r^\pi = V^\pi$, the true value function under π.
Step sizes. Convergence requires the Robbins–Monro conditions to hold *per state*: $\sum_n \alpha_n(s)\mathbf{1}(s_n = s) = \infty$ and $\sum_n \alpha_n(s)^2\mathbf{1}(s_n = s) < \infty$ a.s. for each s.

7.2 Q-learning as asynchronous SA

Q-learning – Asynchronous SA

The update. Q-learning updates a single state-action entry per step:

$$Q_{n+1}(s_n, a_n) = Q_n(s_n, a_n) + \alpha_n(s_n, a_n)\big[R_{n+1} + \gamma \max_{a'} Q_n(s_{n+1}, a') - Q_n(s_n, a_n)\big].$$

SA form. The mean field h is now *nonlinear* due to the max operator:

$$h(Q)(s,a) = r(s,a) + \gamma \sum_{s'} P(s' \mid s,a) \max_{a'} Q(s',a') - Q(s,a).$$

Associated ODE. $\dot{Q} = h(Q)$, which is the Bellman optimality operator $\mathcal{T}^*Q - Q$ in continuous time. Since $\mathcal{T}^*$ is a γ-contraction in the sup-norm, the ODE has Q^* (the optimal Q-function) as its unique globally asymptotically stable equilibrium. To see this, note that the Lyapunov function $V(Q) = \|Q - Q^*\|_\infty$ satisfies

$$\frac{d}{dt}\|Q(t) - Q^*\|_\infty \leq -(1-\gamma)\|Q(t) - Q^*\|_\infty.$$

Complication: asynchronous updates. Q-learning updates only one (s,a) pair per step, making it an *asynchronous* SA scheme. The theory extends via results of Borkar (1998) and Tsitsiklis (1994): if each (s,a) pair is updated infinitely often (ensured by an exploring policy), the asynchronous iteration converges to the same limit as the synchronous ODE.
Conclusion. Under the Robbins–Monro step-size conditions (per state-action pair) and sufficient exploration, Q-learning converges a.s. to Q^*.

7.3 Policy gradient (REINFORCE) as SA

REINFORCE as SA on the Gradient Ascent ODE

The update. REINFORCE updates the policy parameters θ by

$$\theta_{n+1} = \theta_n + \alpha_n \widehat{\nabla J(\theta_n)},$$

where $\widehat{\nabla J(\theta_n)}$ is an unbiased estimate of the policy gradient $\nabla_\theta J(\theta)$.
SA form. This is Robbins–Monro with $h(\theta) = \nabla J(\theta)$ and noise $M_{n+1} = \widehat{\nabla J(\theta_n)} - \nabla J(\theta_n)$.
Associated ODE. $\dot{\theta} = \nabla J(\theta)$, which is gradient ascent on the expected return. Any local maximum θ^* of J with $\nabla J(\theta^*) = 0$ and negative definite Hessian is a stable equilibrium.
Complication. The ODE $\dot{\theta} = \nabla J(\theta)$ typically has multiple stable equilibria (local maxima of J). By Proposition 5.2, the SA iterates converge to one of them, but global optimality is not guaranteed.
Convergence rate. With proper step sizes, the CLT (Theorem 4.1) applies near each local maximum, giving $\sqrt{n}(\theta_n - \theta^*) \xrightarrow{d} \mathcal{N}(0, V)$ for an appropriate covariance V. The high variance of REINFORCE estimators means Σ (and hence V) is large, motivating the use of baselines.

7.4 Actor-critic as two-timescale SA

Actor-Critic – Two-Timescale Analysis

Setup. Actor-critic maintains two parameter vectors:

- Critic parameters $w \in \mathbb{R}^{d_w}$ (fast timescale, step size b_n)
- Actor parameters $\theta \in \mathbb{R}^{d_\theta}$ (slow timescale, step size a_n)

with $a_n/b_n \to 0$.

Coupled iterations.

$$w_{n+1} = w_n + b_n\big[\delta_n \phi(s_n)\big], \quad \delta_n = R_{n+1} + \gamma\phi(s_{n+1})^T w_n - \phi(s_n)^T w_n,$$
$$\theta_{n+1} = \theta_n + a_n\big[\delta_n \nabla_\theta \log \pi_\theta(a_n \mid s_n)\big].$$

Mapping to two-timescale SA.

- $y_n = w_n$, $x_n = \theta_n$.
- $g(\theta, w) = \mathbb{E}_{\pi_\theta}[\delta_w^\theta \phi(s)]$: the critic drift, which is exactly the linear SA drift for TD(0) under policy π_θ.
- $f(\theta, w) = \mathbb{E}_{\pi_\theta}[\delta_w^\theta \nabla_\theta \log \pi_\theta(a \mid s)]$: the actor drift, which approximates $\nabla J(\theta)$ when w encodes V^{π_θ}.

Verifying assumptions.

1. (E5) (fast equilibrium): For fixed θ, the critic ODE $\dot{w} = g(\theta, w)$ is linear SA for TD under π_θ. Its equilibrium $\lambda(\theta) = w^*(\theta)$ is the TD fixed point (projection of V^{π_θ} onto the linear space).
2. (E6) (slow stability): With $w = \lambda(\theta)$, the actor ODE becomes $\dot{\theta} = \nabla J(\theta)$ (approximately), which ascends the performance surface.

Conclusion. By Theorem 5.1, the critic converges to the TD fixed point $w^*(\theta^*)$ and the actor converges to a (local) optimum θ^* of $J(\theta)$.

8 Exercises

Exercise 8.1 (Step-size conditions)**.** For each of the following step-size sequences, determine whether both Robbins–Monro conditions ($\sum a_n = \infty$, $\sum a_n^2 < \infty$) are satisfied:

(a) $a_n = 1/n$

(b) $a_n = 1/n^2$

(c) $a_n = 1/\sqrt{n}$

(d) $a_n = c/(n+d)$ for constants $c, d > 0$

(e) $a_n = 1/(n \log n)$ for $n \geq 2$

(f) $a_n = 1/n^{3/4}$

For each case that fails, explain which condition fails and the practical consequence for an SA algorithm using that step size.

Exercise 8.2 (Scalar Robbins–Monro)**.** Consider the scalar SA iteration $x_{n+1} = x_n + a_n(c - x_n + \xi_{n+1})$ where $c \in \mathbb{R}$ and $\{\xi_n\}$ are i.i.d. with $\mathbb{E}[\xi_n] = 0$ and $\mathbb{E}[\xi_n^2] = \sigma^2 < \infty$.

(a) Identify $h(x)$ and x^*.

(b) Verify the monotonicity condition (A4).

(c) Show that $A = Dh(x^*) = -1$ (the Jacobian at the equilibrium).

(d) With $a_n = a/n$, compute the asymptotic covariance $V = \lim_{n\to\infty} n\mathbb{E}[(x_n - x^*)^2]$ using the Lyapunov equation (8).

(e) Find the optimal a that minimizes V, and compute V_{opt}.

Exercise 8.3 (ODE for Q-learning)**.** Consider a two-state, two-action MDP with $\mathcal{S} = \{1, 2\}$, $\mathcal{A} = \{L, R\}$, $\gamma = 0.9$, deterministic transitions $P(2 \mid 1, L) = 1$, $P(1 \mid 1, R) = 1$, $P(1 \mid 2, L) = 1$, $P(2 \mid 2, R) = 1$, and rewards $r(1, L) = 1$, $r(1, R) = 0$, $r(2, L) = 0$, $r(2, R) = 2$.

(a) Compute $Q^*(s, a)$ for all (s, a) pairs by solving the Bellman optimality equation.

(b) Write out the associated ODE $\dot{Q} = h(Q)$ explicitly (as a 4-dimensional system).

(c) Verify that Q^* is the unique equilibrium.

(d) *(Bonus)* Simulate the ODE numerically from several initial conditions and verify convergence to Q^*.

Exercise 8.4 (Lyapunov function for contraction)**.** Let $T : \mathbb{R}^d \to \mathbb{R}^d$ be a γ-contraction in the sup-norm with fixed point x^*: $\|Tx - Tx'\|_\infty \le \gamma\|x - x'\|_\infty$. Define the ODE $\dot{x} = Tx - x$.

(a) Show that $V(x) = \|x - x^*\|_\infty$ is a Lyapunov function satisfying $\dot{V} \le -(1 - \gamma)V$.

(b) Conclude that x^* is globally asymptotically stable with exponential convergence rate $1-\gamma$.

(c) Apply this to the Bellman optimality operator $T = \mathcal{T}^*$ to verify the ODE analysis for Q-learning.

Exercise 8.5 (Two-timescale separation)**.** Consider the two-timescale system in $\mathbb{R}^2$:

$$\begin{aligned} x_{n+1} &= x_n + a_n(-x_n + y_n + M_{n+1}^{(1)}), \\ y_{n+1} &= y_n + b_n(-2y_n + x_n + M_{n+1}^{(2)}), \end{aligned}$$

where $a_n = 1/n$, $b_n = 1/n^{2/3}$, and the noises are i.i.d. with zero mean and unit variance.

(a) Verify the timescale separation condition $a_n/b_n \to 0$.

(b) For the fast timescale: fix x and find the equilibrium $\lambda(x)$ of $\dot{y} = -2y + x$.

(c) For the slow timescale: substitute $y = \lambda(x)$ into the x-ODE and find the equilibrium x^*.

(d) Find (x^*, y^*) and verify that the coupled ODE system has this as its unique equilibrium.

(e) Verify the Hurwitz conditions for both the fast and slow ODEs.

Exercise 8.6 (Polyak–Ruppert for mean estimation)**.** Let $X_1, X_2, \ldots$ be i.i.d. with $\mathbb{E}[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2$. Consider the SA iteration $\theta_{n+1} = \theta_n + a_n(X_{n+1} - \theta_n)$ with $a_n = 1/n^{2/3}$.

(a) Show this is SA with $h(\theta) = \mu - \theta$ and identify A and Σ.

(b) What is the asymptotic distribution of $\sqrt{n}(\theta_n - \mu)$? (Use the CLT for SA, Theorem 4.1, noting the step size is a/n^α with $\alpha = 2/3$.)

(c) Define $\bar{\theta}_n = \frac{1}{n}\sum_{k=1}^n \theta_k$. What is the asymptotic distribution of $\sqrt{n}(\bar{\theta}_n - \mu)$?

(d) Compare the asymptotic variances from (b) and (c). Which is better?

(e) Compare both to the sample mean $\bar{X}_n = \frac{1}{n}\sum_{k=1}^n X_k$. What is the asymptotic variance of $\sqrt{n}(\bar{X}_n - \mu)$?

Exercise 8.7 (Hurwitz verification for TD with linear FA)**.** Consider a three-state MDP with $\mathcal{S} = \{1, 2, 3\}$, policy π inducing transition matrix

$$P^\pi = \begin{pmatrix} 0.2 & 0.5 & 0.3 \\ 0.1 & 0.6 & 0.3 \\ 0.4 & 0.3 & 0.3 \end{pmatrix},$$

discount $\gamma = 0.95$, and stationary distribution $d^\pi = (0.2, 0.45, 0.35)$. Let the feature matrix be $\Phi = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$.

(a) Compute the matrix $A = \Phi^T D(\gamma P^\pi - I)\Phi$ where $D = \mathrm{diag}(d^\pi)$.

(b) Verify that A is negative definite (hence Hurwitz).

(c) Conclude that TD(0) converges with this function approximation scheme.

Exercise 8.8 (Convergence failure: the deadly triad)**.** Construct a simple two-state MDP with linear function approximation and off-policy sampling such that the matrix A in the linear SA formulation of TD(0) has an eigenvalue with positive real part, demonstrating divergence. Specifically:

(a) Take $\mathcal{S} = \{1, 2\}$, $\gamma = 0.99$, and a behavior policy that visits state 1 with probability 0.01 and state 2 with probability 0.99.

(b) Let P^π be the transition matrix under the target policy π, with $P^\pi(1 \mid 1) = 0.9$, $P^\pi(2 \mid 1) = 0.1$, $P^\pi(1 \mid 2) = 0.1$, $P^\pi(2 \mid 2) = 0.9$.

(c) Use the feature representation $\phi(1) = 1$, $\phi(2) = 2$ (scalar features).

(d) Compute $A = \mathbb{E}_\mu[\phi(s)(\gamma\phi(s') - \phi(s))]$ where μ is the behavior distribution, and show it has a positive eigenvalue.

(e) Explain why this SA iteration diverges.

Exercise 8.9 (SA with Markovian noise)**.** The standard SA theory assumes martingale difference noise. In RL, the noise is typically *Markovian* (since observations come from a Markov chain). Explain how the ODE method extends to Markovian noise under the following conditions:

(a) The Markov chain $\{Y_n\}$ is ergodic with stationary distribution μ.

(b) $h(x) = \int H(x, y)\, d\mu(y) = \mathbb{E}_\mu[H(x, Y)]$.

(c) The "Poisson equation" $g(x, y) - Pg(x, y) = H(x, y) - h(x)$ has a solution g with suitable regularity.

Sketch how g is used to decompose the noise into a martingale difference term plus a telescoping (bounded) term, recovering the standard ODE method assumptions.

Exercise 8.10 (Nonlinear two-timescale: GTD2)**.** The GTD2 algorithm (Sutton et al., 2009) uses two-timescale SA to perform off-policy TD learning with linear function approximation. The updates are:

$$\begin{aligned} w_{n+1} &= w_n + b_n\big[\phi_n(\phi_n - \gamma\phi'_n)^T\theta_n - \phi_n\phi_n^T w_n\big], \\ \theta_{n+1} &= \theta_n + a_n\big[\delta_n\phi_n - \gamma\phi'_n(\phi_n^T w_n)\big], \end{aligned}$$

where $\phi_n = \phi(s_n)$, $\phi'_n = \phi(s_{n+1})$, $\delta_n = R_{n+1} + \gamma\phi_n'^T\theta_n - \phi_n^T\theta_n$.

(a) Identify the fast and slow iterates and verify the timescale separation.

(b) Compute the fast-timescale equilibrium $\lambda(\theta)$ for the w-update.

(c) Substitute $\lambda(\theta)$ into the θ-update to obtain the slow-timescale ODE.

(d) Show that the slow-timescale ODE is gradient descent on the *mean squared projected Bellman error* (MSPBE).

9 Summary of Key Results

Result	**Key Condition**	**RL Application**
Robbins–Monro (Thm 2.1)	Monotone h, MDS noise, $\sum a_n = \infty$, $\sum a_n^2 < \infty$	SGD convergence for convex losses
ODE Method (Thm 3.2)	Bounded iterates, Lipschitz h, stable ODE equilibrium	Primary tool for all RL convergence proofs
Borkar–Meyn (Thm 3.3)	h_∞ ODE has stable origin	Verifies boundedness for linear SA / TD
SA CLT (Thm 4.1)	$a_n = a/n$, Hurwitz Jacobian	Rate: $\|x_n - x^*\| = O(1/\sqrt{n})$
Polyak–Ruppert (Thm 4.2)	Average iterates, $a_n = O(n^{-\alpha})$, $\alpha \in (1/2, 1)$	Optimal rate via iterate averaging (target networks, EMA)
Two-Timescale (Thm 5.1)	$a_n/b_n \to 0$, fast/slow ODE stability	Actor-critic, GTD, GTD2 convergence
Linear SA (Thm 6.3)	A Hurwitz	TD(0) with linear FA (on-policy)
Lock-in (Prop 5.2)	Multiple stable equilibria, nondegenerate noise	Policy gradient converges to local optimum, avoids saddle points

RL Algorithm	**SA Type**	$h(x)$ / **Drift**	**Key SA Theorem**
TD(0) (tabular)	Async. linear SA	$(\gamma P^\pi - I)V + r^\pi$	ODE Method + Hurwitz
TD(0) (linear FA)	Linear SA	$Aw+b$, A Hurwitz (on-policy)	Thm 6.3
Q-learning	Async. nonlinear SA	$\mathcal{T}^*Q - Q$ (γ-contraction)	ODE + contraction
REINFORCE	SA (gradient ascent)	$\nabla J(\theta)$	ODE + lock-in
Actor-Critic	Two-timescale SA	Critic: TD drift; Actor: policy gradient	Thm 5.1
GTD / GTD2	Two-timescale SA	Gradient of MSPBE	Thm 5.1

The central message of this document is a single principle:

The SA–ODE Correspondence

Every RL update rule of the form $x_{n+1} = x_n + a_n[H(x_n, Y_n)]$ can be analyzed by:

1. Identifying the mean field $h(x) = \mathbb{E}[H(x, Y)]$.
2. Writing the associated ODE $\dot{x} = h(x)$.
3. Analyzing the stability of the ODE (Lyapunov functions, contraction, Hurwitz matrices).
4. Applying the appropriate SA convergence theorem.

If the ODE has a stable equilibrium and the noise/step-size conditions are met, the RL algorithm converges. If the ODE is unstable (e.g., off-policy TD with FA), the RL algorithm diverges.